

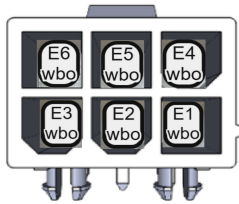


Year
4Runner: 1996
Tacoma/T100: 1995-1997
Engine: 5VZ-FE (3.4L V6)
Board: uaEFI Adapter Board V1.0

Notes:
Manual Transmission Only
Manual Transmission Only

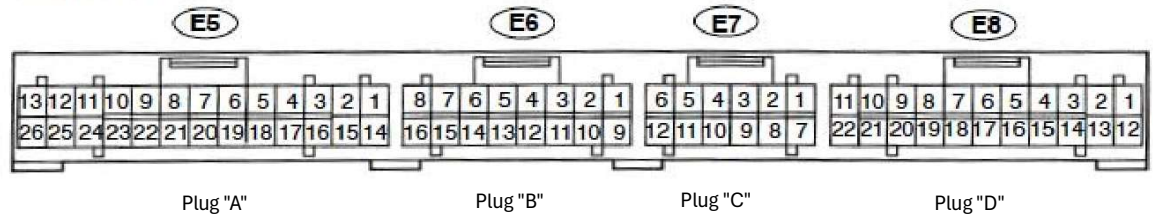
****WARNING**** Do not plug your ECU into your laptop/computer if your laptop/computer is plugged into the vehicle's 12v charging port!

Wideband Controller
"WBO"



Plug "E"

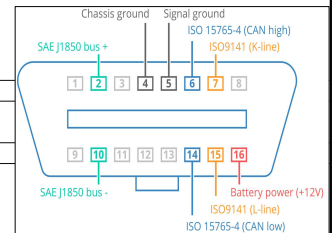
ECM Terminals



OPEN PINS

Note: Populate open slots in ECU connectors with 040IIIU-SKT (OEM p/n 82998-12690)

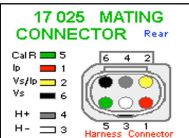
Connector	Plug	Pin	Function	Tuner Studio Channel Name	Notes
E5	A	1	Lowside Output (with PWM)	B17 Low Side Output 3	PWM Ground Output
E5	A	15	Ignition Output 6	B10 Coil 6	**MUST be used with an igniter or smart coils**
E5	A	16	Ignition Output 5	B13 Coil 5	**MUST be used with an igniter or smart coils**
E5	A	20	Ignition Output 4	B11 Coil 4	**MUST be used with an igniter or smart coils**
Connector	Plug	Pin	Function	Tuner Studio Channel Name	Notes
E6	B	15	AUX2	C3 AUX2	Analog Input #2 (0-5v)
E6	B	16	AUX1	D1 AUX1	Analog Input #1 (0-5v)
Connector	Plug	Pin	Function	Tuner Studio Channel Name	Notes
E7	C	1	HB1+		H-Bridge 1 +
E7	C	2	HB1-		H-Bridge 1 -
E7	C	8	5V Sensor		+5V for sensors
E7	C	10	Sensor Ground		For sensor grounds only
Connector	Plug	Pin	Function	Tuner Studio Channel Name	Notes
E8	D	1	+B		Switched +12V power from EFI relay circuit
E8	D	2			
E8	D	3	CAN1H	N/A	CANBUS High If connecting to stock OBD2 port, use pin p/n BD-SKT (OEM p/n 776001-1) and populate OBD2 pin #6
E8	D	5	7th Injector	B9 MR Weak LS 1	Connect 7th injector negative here, and positive to +B (12V EFI Relay Circuit)
E8	D	10	Lowside Output (with PWM)	B8 Low Side Output 2	PWM Ground Output (flyback diode has been added to the adapter board for protection)
E8	D	13	Flex	D5 FLEX	Flex Fuel Signal
E8	D	14	CAN1L	N/A	CANBUS Low If connecting to stock OBD2 port, use pin p/n BD-SKT (OEM p/n 776001-1) and populate OBD2 pin #14
E8	D	15	TPS2	C14 TPS2	Throttle Position Sensor 2 (Can be used for any 0-5v Analog Input)
E8	D	16	PPS2	C4 PPS2	Pedal Position Sensor 2 (Can be used for any 0-5v Analog Input)
E8	D	17	PPS1	D6 PPS1	Pedal Position Sensor 1 (Can be used for any 0-5v Analog Input)
E8	D	18	MAP	D9 MAP	Manifold Absolute Pressure Sensor (Can be used for any 0-5v Analog Input)
E8	D	21	Digital Input	MEGA_EPIC_D31	Ground switched
E8	D	22	Digital Input	MEGA_EPIC_D30	Ground switched



Note:

Use mating connector Molex 0039012060 or 0039012065
Populate with Molex Mini-Fit Jr 5556 series pins (p/n 0039000038 for 18-24AWG wire, p/n 0039000077 for 16AWG wire)
This connector is to be used with a Bosch LSU 4.9 type sensor (p/n 17025)
The mating connector to the Bosch LSU 4.9 sensor is Bosch BAK-6S p/n 1 928 404 669 or 1 987 280 016

Connector	Plug	Pin	Function	Wire Color	Notes
WBO	E	1	H+	Grey	Heater positive
WBO	E	2	CalR	Green	Factory calibration / trim resistor
WBO	E	3	H-	White	Heater negative
WBO	E	4	Vs	Black	Nernst-cell voltage
WBO	E	5	Ip	Red	Pump-cell current
WBO	E	6	Vs/Ip	Yellow	Virtual ground / reference for pump cell





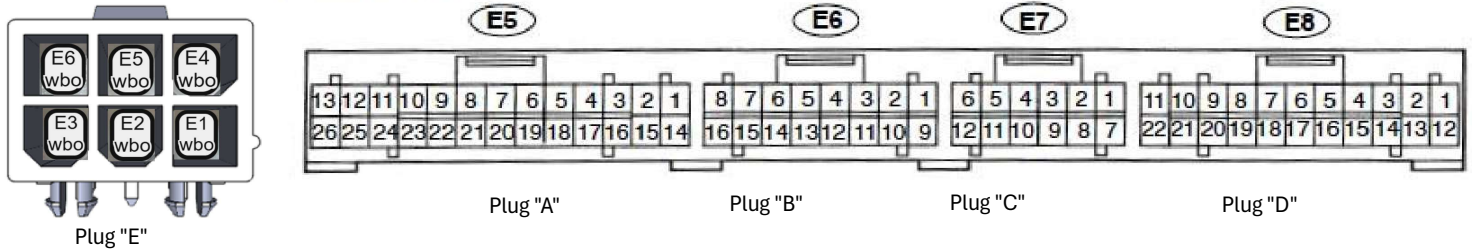
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Wideband Controller
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ECM Terminals



POPULATED PINS

Note: Populate open slots in ECU connectors with 040IIIU-SKT (OEM p/n 82998-12690)

Connector	Plug	Pin	Function	Tuner Studio Channel Name	Notes
E5	A	2	ACV	MEGA_EPIC_D45	On/Off Ground Output (this is connected to the A/C Idle Up valve on some Tacoma & T100)
E5	A	4	IDL	MEGA_EPIC_D40	Idle Sensor
E5	A	5	EVP	MEGA_EPIC_D3	EVAP Solenoid
E5	A	6	RSC	B18 Low Side Output 2	Idle Air Control Valve - Close
E5	A	7	RSO	B7 Low Side Output 1	Idle Air Control Valve - Open
E5	A	8	#60	B1 Injector Output 6	Injector #6
E5	A	9	#50	B2 Injector Output 2	Injector #5
E5	A	10	#40	B3 Injector Output 4	Injector #4
E5	A	11	#20	B5 Injector Output 2	Injector #2
E5	A	12	#10	B6 Injector Output 1	Injector #1
E5	A	14	FC	B16 Low Side Output 4	Fuel Pump
E5	A	18	EGR	MEGA_EPIC_D44	On/Off Ground Output (this is connected to the EGR on some Tacoma & T100)
E5	A	19	FPU	MEGA_EPIC_D46	On/Off Ground Output (this is connected to the Fuel Pressure Up valve on some Tacoma & T100)
E5	A	21	IGT3	B12 Coil 3	Ignition Coil #3
E5	A	22	IGT2	B14 Coil 2	Ignition Coil #2
E5	A	23	IGT1	B15 Coil 1	Ignition Coil #1
E5	A	25	#30	B4 Injector Output 3	Injector #3
Connector	Plug	Pin	Function	Tuner Studio Channel Name	Notes
E6	B	2	VG	C15 AUX3	Mass Air Flow Sensor Signal
E6	B	3	KNK2	D14 Knock Input	Knock Sensor Bank 2 (use selector switch on carrier board to choose KNK1 or KNK2)
E6	B	4	THW	D16 CLT Coolant	Engine Coolant Temperature
E6	B	5	OX1	MEGA_EPIC_A1	Upstream Narrowband Oxygen Sensor Signal
E6	B	6	KNK1	D14 Knock Input	Knock Sensor Bank 1 (use selector switch on carrier board to choose KNK1 or KNK2)
E6	B	10	VTA (TPS)	D13 TPS1	Throttle Position Sensor
E6	B	11	IDL	MEGA_EPIC_D40	Idle Sensor
E6	B	12	THA	D15 IAT	Intake Air Temperature
E6	B	13	OX2	MEGA_EPIC_A15	Downstream Narrowband Oxygen Sensor Signal
Connector	Plug	Pin	Function	Tuner Studio Channel Name	Notes
E7	C	3	HT1	MEGA_EPIC_D13	Upstream O2 Sensor Heater
E7	C	4	PTNK	MEGA_EPIC_ADC_14	EVAP Pressure
E7	C	9	HT2	MEGA_EPIC_D12	Downstream Oxygen Sensor Heater
E7	C	11	G	C7 HALL3	Cam Signal
E7	C	12	NE	C6 HALL2	Crank Signal
Connector	Plug	Pin	Function	Tuner Studio Channel Name	Notes
E8	D	4	W	MEGA_EPIC_D43	Check Engine Light
E8	D	6	ACT	MEGA_EPIC_D49	A/C Terminate
E8	D	7	AC1	MEGA_EPIC_D23	A/C Request
E8	D	8	SP1	C5 HALL1	Speed Sensor 1
E8	D	9	4WD	MEGA_EPIC_D29	4WD Indicator
E8	D	11	STA	MEGA_EPIC_D28	Cranking Signal
E8	D	20	BK	MEGA_EPIC_D22	Brake Light